

HARVEST-VISION

On the Evaluation of Multimodal Large Language Models for Agricultural Image Classification across Diverse Tasks

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Presented by

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MOTIVATIONS

- Plant identification is crucial for agriculture and **food security**
- Early disease detection prevents yield loss
- Manual inspection is:
 - **labor intensive**
 - **error-prone**
 - **limited scalability**

**An automated system is required which enables
fast and reliable crop monitoring.**

EXAMPLES



Cashew



Cassava



Bacterial Blight



Brown Spot



Maize



Tomato



Green Mite



Mosaic

(a) Healthy leaves

(b) Cassava leaf diseases

Figure: Representative categories of plant leaf images

OPPORTUNITIES

- Advances in **computer vision** and **machine learning**
- CNNs can learn leaf patterns and disease symptoms
 - require large labeled datasets and **task-specific training**
- LLMs: **strong reasoning** and contextual understanding
- Multimodal LLMs : **visual perception** + language reasoning

**Potential to build scalable, intelligent
crop monitoring systems**

RESEARCH QUESTIONS

- Can MLLMs **classify plant species** from leaf images?
- Can they **detect diseases** from leaf symptoms?
- Can they **recognize crops** without predefined taxonomy?
- How do open-source vs proprietary models compare?
- How do the results vary across different datasets?

We seek to understand the strengths and limitations of multimodal LLMs in agricultural vision tasks.

CONTRIBUTIONS

- ✓ **Multimodal LLMs to classify plant species from leaf images**
 - **Both zero-shot and few-shot prompting strategies**
- ✓ **Multimodal LLMs in detecting and distinguishing plant diseases**
 - **Few-shot prompting enhancing the performance**
- ✓ **Extend classification tasks to multiple plant parts**
 - **Leaves, fruits, flowers, stems, seeds, and full plants**
 - **Open-vocabulary crop recognition without fixed taxonomies**
- ✓ **We utilize both open-source and black-box models**
 - **Generalize across datasets, plant types and disease categories**

MULTIMODAL LARGE LANGUAGE MODELS

- ✓ **Multimodal LLMs jointly process and reason over text and images**
 - **integrate visual input with language understanding**
- ✓ **Black-Box Proprietary Models:**
 - **Examples: GPT-5, Gemini 3 etc.**
 - **Performance across tasks makes a suitable benchmark**
 - **GPT-4o and GPT-5 are accessed through APIs**
- ✓ **Open-source MLLMs**
 - **Examples: Gemma3-4B, LLaVA-v1.5-7B and Qwen2.5-7B**
 - **Their open architectures enable controlled adaptation**

PLANT SPECIES CLASSIFICATION

You are an AI expert at leaf-species identification. Given an image of a crop leaf, determine whether it belongs to one of these four species: Cashew, Cassava, Maize, or Tomato. Then output EXACTLY:

Conclusion: one of the above labels.

Zero-shot



Few-shot



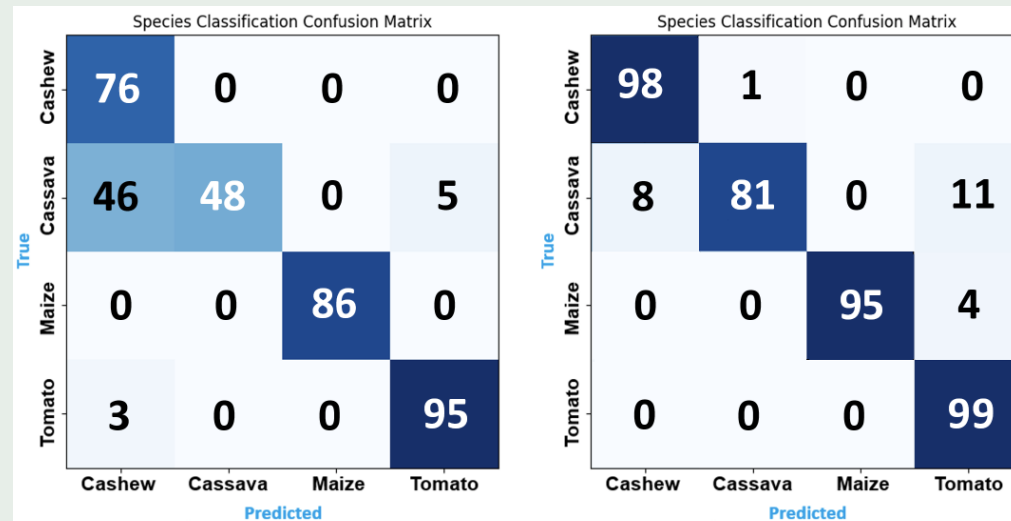
You are an AI expert at leaf-species identification. Use the examples above to learn each species' visual cues. Now, for the given test image, determine whether it belongs to one of these four species: Cashew, Cassava, Maize, or Tomato. Then output EXACTLY:

Conclusion: one of the above labels.

RESULTS ON PLANT SPECIES CLASSIFICATION

Table: Performance comparison among zero-shot and few-shot methods on plant-species classification.

LLM	Zero-shot		Few-shot	
	Accuracy	F1 Score	Accuracy	F1 Score
GPT-5	82.25%	0.85	93.50%	0.94
GPT-4	76.25%	0.84	93.25%	0.94
Gemma3	38.0%	0.34	47.75%	0.38



CROP DISEASE CLASSIFICATION

Leaves	Cashew	Cassava	Maize	Tomato
Diseases	Anthracnose, Gumosis, Leaf miner, Red rust	Bacterial blight, Brown spot, Green mite, Mosaic	Fall armyworm, Grasshopper, Leaf beetle, Leaf blight, Leaf spot, Streak virus	Leaf blight, Leaf curl, Septoria leaf spot, Verticillium wilt

You are an AI specialist in Cashew leaf-disease diagnosis. Given an image of a cashew leaf, classify it as one of these categories: Cashew healthy, Cashew anthracnose, Cashew gumosis, Cashew leaf miner, and Cashew red rust. Then output EXACTLY:

Conclusion: one of the above labels.

Zero-shot



RESULTS ON CROP DISEASE CLASSIFICATION

Table: Performance comparison among zero-shot and few-shot methods on crop disease classification using GPT models.

Crop	Zero-shot		Few-shot	
	GPT-4o	GPT-5	GPT-4o	GPT-5
Cashew	60%	67%	87%	78%
Cassava	64%	80%	74%	81%
Maize	81%	81%	82%	82%
Tomato	50%	50%	54%	52%

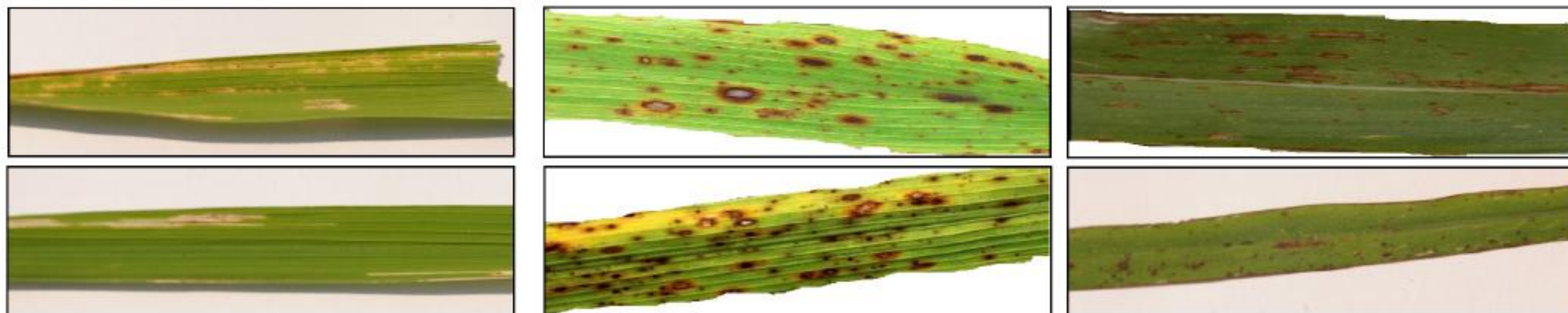
BB	2	17	1	0	0
BS	0	19	0	1	0
GM	0	0	6	6	8
Healthy	0	0	0	20	0
Mosaic	0	1	2	0	17
	BB	BS	GM	Healthy	Mosaic

(a) zero-shot

BB	10	9	0	0	1
BS	0	19	0	1	0
GM	0	0	5	3	12
Healthy	0	0	0	20	0
Mosaic	0	0	0	0	20
	BB	BS	GM	Healthy	Mosaic

(b) few-shot

CROP DISEASE CLASSIFICATION : NEW DATASET



(a) Bacterial leaf blight

(b) Brown Spot

(c) Leaf Smut

Figure: Different diseases of rice leaves which include (a) Bacterial leaf blight, (b) Brown Spot, and (c) Leaf Smut

LLM	Zero-shot		Few-shot	
	Acc.	F1 Score	Acc.	F1 Score
GPT-5	65.56%	0.55	93.33%	0.93
GPT-4	65.56%	0.55	90.00%	0.89
Gemma3	27.78%	0.22	34.44%	0.22
LLaVA-v1.5	36.67%	0.35	38.89%	0.39
Qwen2.5	33.33%	0.17	33.33%	0.17

Table: Classifying rice leaf diseases using different black-box and open-source LLMs in terms of accuracy and F1 scores

TAXONOMY-FREE CROP CLASSIFICATION

You are an expert in agricultural crop identification. Your task is: given an image of a crop's tree, plant, fruit, flower, leaf, seed, or field, output the most likely crop name. The response should consist of exactly one concise label, such as 'mango' or 'rose'. If the crop cannot be confidently identified, the output should be 'Unknown'. To ensure consistency, avoid providing extra descriptions, synonyms, explanations, or multiple options. Then output EXACTLY:

Conclusion: "crop name" or "Unknown".



Zero-shot

EXAMPLES



(a) Banana



(b) Sugarcane



(c) Cherry

- ✓ This task is particularly challenging for two main reasons:
 - **Taxonomy-free: no fixed set of predefined labels**
 - **Not limited to leaves: can include fruits or entire crop fields**

RESULTS

Crops Names		
Cherry	Coffee-plant	Cucumber
Lemon	Tobacco-plant	Almond
Banana	Pearl-millet	Coconut
Gram	Mustard-oil	Sunflower
Jute	Vigna-radiati	Tea
Wheat	Pineapple	Clove
Maize	Rice	Jowar
Sugarcane	Soyabean	Cotton
Papaya	Olive-tree	Cardamom
Tomato	Fox-nut	Chilli

Metric	Zero-shot		Few-shot	
	GPT-4	GPT-5	GPT-4	GPT-5
Accuracy	48.53%	51.33%	52.83%	56.17%

Table: Performance Comparison of GPT-4o and GPT-5 on Taxonomy-free Agricultural Crop Recognition

- ✓ **These accuracies are promising for a **challenging 30-way** taxonomy-free classification task.**
- ✓ **However, they also highlight **considerable room for improvement** in handling such diverse crop imagery.**

DISCUSSIONS

- **Potential of LLMs for plant species and disease classification**
 - **Both zero-shot and few-shot across diverse plant datasets**
 - **Challenging taxonomy-free crop classification task**
- **Zero-shot evaluations relies solely on pretraining and prior exposure**
- **Few-shot helped models better adapt to fine-grained recognition**
- **Proprietary models performed better than the open-source models**
- **Accuracy remains modest in the taxonomy-free setting**

FUTURE WORK

- ✓ **Domain adaptation for agriculture**
 - **Fine-tuning multimodal models on plant-specific datasets**
 - **Incorporating expert plant pathology annotations**
- ✓ **Improving contextual understanding**
 - **Integrating retrieval-augmented generation (RAG)**
 - **Enhancing model interpretability and reasoning**
- ✓ **Handling fine-grained plant diseases**
 - **Addressing visually similar disease symptoms**
 - **Improving detection of rare and emerging plant diseases**

More accurate, trustworthy, and scalable real-world agricultural decision support system

THANK YOU!

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